

EXP.NO: 3	Write a Servlet program that prints "Hello, World!" when accessed through a browser.
DATE:20/02/2025	

AIM:

To write a basic Servlet program that demonstrates the use of Java Servlets by printing "Hello, World!" when accessed through a browser.

ALGORITHM:

1. Input:

- The user makes an HTTP request (usually by entering a URL) through a browser.

2. Servlet Program Flow:

1. Create a Servlet Class:

- Define a Java class that extends the `HttpServlet` class.
- Override the `doGet()` method, which handles HTTP GET requests from the client (browser).

2. Handle HTTP Request:

- In the `doGet()` method:
 - Get the **PrintWriter** object to write the response to the browser.
 - Set the response content type to **text/html**.
 - Write the **"Hello, World!"** message to the response.

3. Compile the Servlet Class:

- Compile the Java Servlet file using the `javac` command to generate a `.class` file.

4. Deploy the Servlet:

- Place the compiled servlet class file in the appropriate directory of the web application (typically in the /WEB-INF/classes folder).
- Configure the servlet in the web.xml file of the web application or use annotations.

5. Access the Servlet via Browser:

- Start the **Apache Tomcat** or another servlet container.
- Access the servlet by entering the appropriate URL (e.g., <http://localhost:8080/your-webapp/hello>).

CODE:

student.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Student System</title>
  <link rel="stylesheet" href="css/style.css">
</head>
<body>

<div class="container">
  <h2>Student Information Management System</h2>

  <a href="student">
    <button>Enter System</button>
  </a>
</div>

</body>
</html>
```

Style.css:

```
body {  
    margin: 0;  
    padding: 0;  
    font-family: Arial, sans-serif;  
    background: linear-gradient(to right, #74ebd5, #ACB6E5);  
    height: 100vh;  
    display: flex;  
    justify-content: center;  
    align-items: center;  
}
```

```
.container {  
    background: white;  
    width: 500px;  
    padding: 35px;  
    border-radius: 12px;  
    text-align: center;  
    box-shadow: 0px 8px 20px rgba(0,0,0,0.2);  
}
```

```
h1 {  
    color: #0d47a1;  
}
```

```
h2 {  
    color: #003366;  
    margin-bottom: 20px;  
}
```

```
h3 {  
    color: green;  
}
```

```
p {  
    color: #555;  
    font-size: 16px;  
}
```

```
button {  
    padding: 12px 22px;  
    font-size: 16px;  
    background: #0d47a1;  
    color: white;  
    border: none;  
    border-radius: 6px;  
    cursor: pointer;  
}
```

```
button:hover {  
    background: #08306b;  
}
```

StudentServlet.java:

```
import java.io.IOException;  
import java.io.PrintWriter;  
import jakarta.servlet.ServletException;  
import jakarta.servlet.annotation.WebServlet;  
import jakarta.servlet.http.HttpServlet;  
import jakarta.servlet.http.HttpServletRequest;  
import jakarta.servlet.http.HttpServletResponse;
```

```

@WebServlet("/student")
public class StudentServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Student System</title>");
        out.println("<link rel='stylesheet' href='" +
            request.getContextPath() + "/css/style.css'>");
        out.println("</head>");

        out.println("<body>");
        out.println("<div class='container'>");

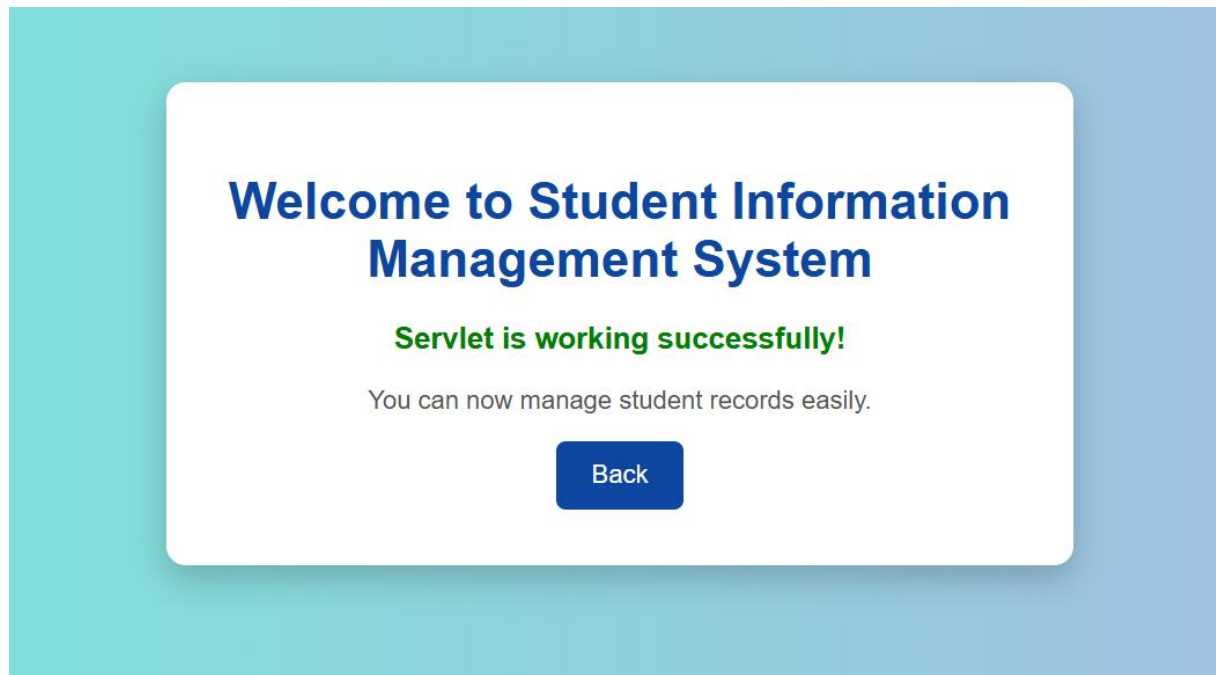
        out.println("<h1>Welcome to Student Information Management
System</h1>");
        out.println("<h3>Servlet is working successfully!</h3>");
        out.println("<p>You can now manage student records
easily.</p>");

        out.println("<a href='student.html'><button>Back</button></a>");

        out.println("</div>");
        out.println("</body>");
        out.println("</html>");
    }
}

```

Output:



RESULT:

The browser will display the message "Hello, World!" returned by the Servlet.